TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 8

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- read each question **carefully**
- answer **all** questions
- write your answers **clearly** in the spaces provided
- **check** your answers.

Information:

- this examination has **two** sections. These are clearly labelled
- you are **not allowed** to use a calculator for Section A
- you **are allowed** to use a calculator for Section B
- the **maximum** mark for this examination is 60
- the marks available for each question are shown in **bold** beneath each question.

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

· Assessment: Past Paper 8 (Non-Calculator & Calculator)

· Complete stepped calculations, methods & mark scheme codes [MI/AI]

· Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)

· Pass standard: 34 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 5.817 – 2.75

Step 1: Align decimals with padding zero:

5.817 - 2.750 = 3.067

Answer 3.067 ✓ *[1 mark]*

[1]

2. Work out 4 × 3.145

Step 1: Multiply whole numbers ignoring decimals:

4 × 3145 = 12 580

Step 2: Place 3 decimal digits: 12.580 = 12.58

Answer 12.58 ✓ *[1 mark]*

[1]

Past paper

Questions continue on the following page

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3. Convert 12 kilometres into miles.

Use 1.6 kilometres = 1 mile

Step 1: Conversion factor: 1.6 km = 1 mile

Miles = 12 ÷ 1.6 = 120 ÷ 16 = 7.5 miles

Answer 7.5 [2 marks] **miles**

[2]

4. There are 824 students at a school.

112 of them are in their final year.

One student is chosen at random.

What is the probability that this student is in their final year?

Give your answer as a fraction in its simplest form.

Step 1: Write probability as fraction: 112 / 824

Step 2: Simplify by dividing numerator & denominator by 8:

112 ÷ 8 = 14 and 824 ÷ 8 = 103

Simplest form = 14 / 103

Answer 14 / 103 [2 marks]

[2]

Past paper

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5. Morgan is the manager of a concert arena.

The number of tickets sold in two years is shown:

Year 1:	**Year 2**:
784 693 tickets	523 154 tickets

Morgan thinks that the total number of tickets sold in these two years is more than 1 400 000

Is Morgan correct?

Show how you decide.

Step 1: Total tickets sold over two years:

Year 1 + Year 2 = 784 693 + 523 154 = 1 307 847 tickets

Step 2: Compare with 1.4 million (1 400 000):

1 307 847 < 1 400 000. Therefore Morgan is not correct.

Answer No (1 307 847 < 1.4m ✓ *[2 marks]*

[2]

Past paper

Questions continue on the following page

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6. Work out the original cost of this television:

Television
20% off
Now only £240

Step 1: £240 represents (100% - 20%) = 80% of original:

Original cost = £240 ÷ 0.80 = £300

Answer £ 300 [2 marks]

[2]

7. Work out:

$$8\frac{1}{4} + 6\frac{1}{5}$$

Give your answer as a mixed number.

Step 1: Add whole numbers: 8 + 6 = 14

Step 2: Common denominator for 4 and 5 is 20:

1/4 = 5/20 and 1/5 = 4/20

Step 3: Add fractions: 5/20 + 4/20 = 9/20

Total = 14 9/20

Answer 14 9/20 [2 marks]

[2]

Past paper

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8. Dua is going to make a gold bracelet using 50 cubic centimetres (cm^3) of liquid gold.

Dua knows the density of gold is 19.3 g/cm^3.

How many grams of gold will Dua need to make the bracelet?

Use: $\text{Density} = \dfrac{\text{mass}}{\text{volume}}$

Step 1: Rearrange formula: Mass = Density × Volume

Density = 19.3 g/cm³, Volume = 50 cm³

Step 2: Calculate mass:

Mass = 19.3 × 50 = 965 grams

Answer 965 ✓ [3 marks] **grams**

[3]

End of Section A.

Past paper

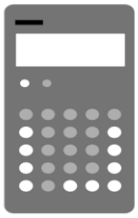
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Section B begins on Page 7.

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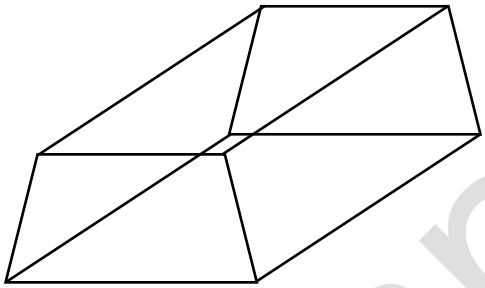
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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. What is this shape?



a) Pyramid	**b)** Trapezium
c) Cuboid	**d)** Prism

Answer	d) Prism ✓ [1 mark]

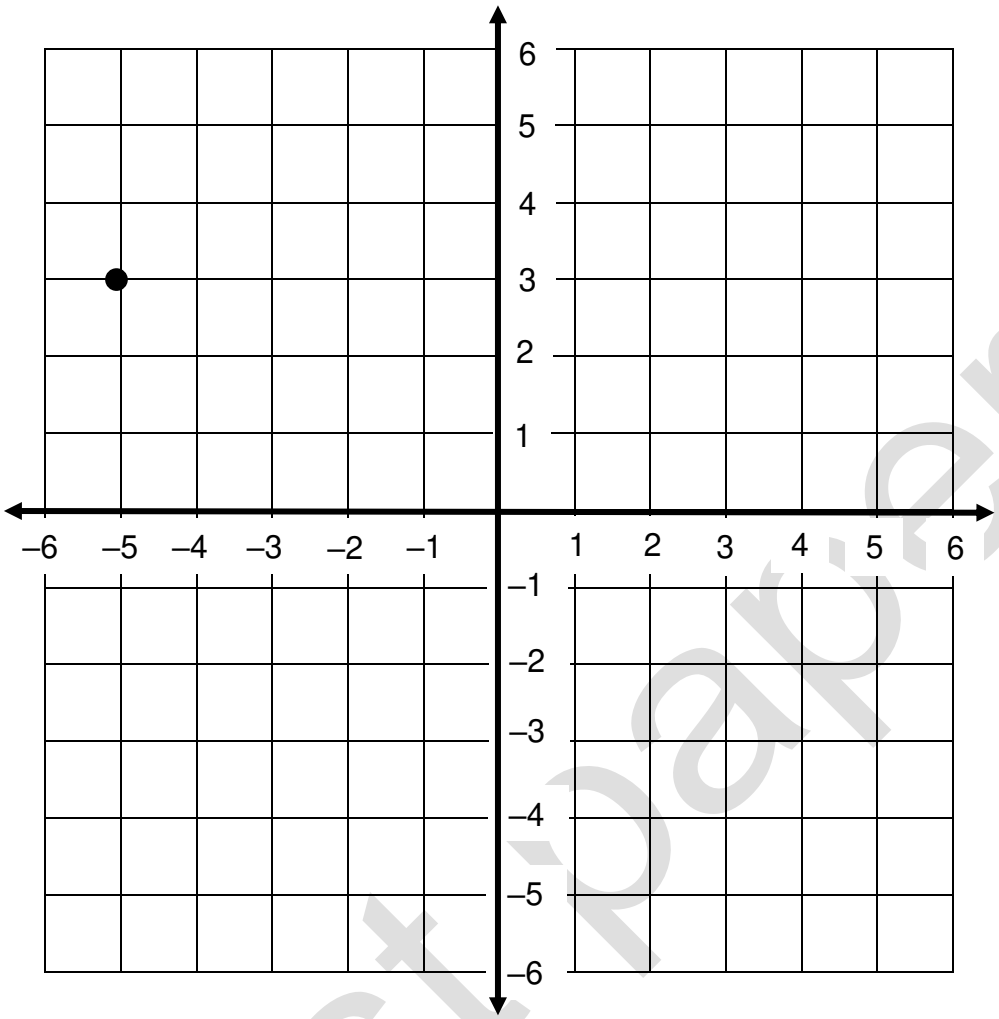
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Past Paper

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2. Write down the coordinates of the point plotted on the grid.



Answer	(-5 , 3)

[1 mark]

[1]

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3. Zola wants a new sofa.

Zola can choose either a grey or blue sofa.

Both sofas have the same original price.

Grey sofa	Blue sofa
$\frac{1}{9}$ off	12% off

Which sofa has the bigger discount?

Show how you decide.

Step 1: Convert Grey sofa discount (1/9) to percentage:

Grey: 1 ÷ 9 = 0.1111... ≈ 11.1%

Step 2: Blue sofa discount is given as 12% (0.12)

Step 3: Compare: 12% > 11.1%, so Blue sofa has bigger discount.

Answer Blue (12% > 11 [2 marks]

[2]

Past paper

Questions continue on the following page

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4. Work out:

$$\frac{4^2 + 40 \div 2}{15 - 3 \times 4}$$

Show your working.

Step 1: Numerator using BIDMAS (Indices, then Division, then Addition):

$4^2 = 16$ and $40 \div 2 = 20$ --> $16 + 20 = 36$

Step 2: Denominator using BIDMAS (Multiplication, then Subtraction):

$3 \times 4 = 12$ --> $15 - 12 = 3$

Step 3: Divide numerator by denominator: $36 \div 3 = 12$

Answer	12 ✓ [2 marks]

[2]

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5. A local farm sells milk in glass bottles for £0.80 per pint.

The farm increases the price to £1.00 per pint.

Work out the percentage increase in price.

Step 1: Increase amount = £1.00 - £0.80 = £0.20

Step 2: Percentage increase = (0.20 ÷ 0.80) × 100% = 25%

Answer	25 ✓ [2 marks] %

[2]

6. Tai needs their house painted within 10 days.

It will take 3 painters 6 days to paint the house.

Will 2 painters working at the same rate get the house painted within 10 days?

Show how you decide.

Step 1: Total workload = 3 painters × 6 days = 18 painter-days

Step 2: Time for 2 painters = 18 ÷ 2 = 9 days

Step 3: Compare with 10 days requirement: 9 days <= 10 days. Yes.

Answer	Yes (takes 9 days) ✓ [2 marks]

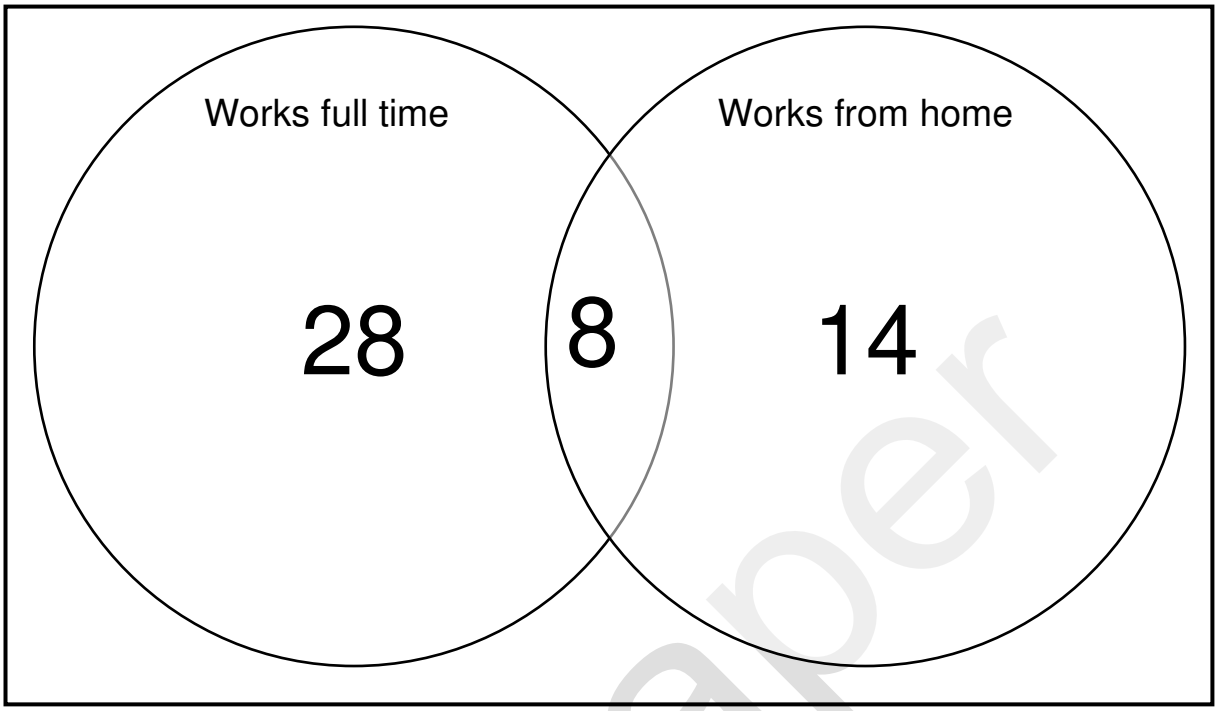
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Questions continue on the following page

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7. The Venn diagram shows the working patterns of a group of employees.



One of these employees is chosen at random.

What is the probability that this employee works full time?

Give your answer as a decimal **and** a percentage.

Step 1: Number of employees working full time = 28 + 8 = 36

Step 2: Total number of employees = 28 + 8 + 14 = 50

Step 3: Probability = 36 ÷ 50 = 72 ÷ 100

Decimal = 0.72 | Percentage = 72%

Answer	a) Decimal: **0.72**	b) Percentage: **72%**

[3 marks]

[3]

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8. Harper walks 15 kilometres in 200 minutes.

Harper thinks this is 4.2 kilometres per hour (kph).

Is Harper correct?

Show how you decide.

Step 1: Convert 200 minutes into hours: 200 ÷ 60 = 3.333... hours

Step 2: Speed = Distance ÷ Time = 15 ÷ (200/60) = 4.5 kph

Step 3: Compare with Harper's estimate of 4.2 kph:

4.5 kph ≠ 4.2 kph (walked at 4.5 kph). Therefore No.

Answer	No (4.5 kph > 4.2 kph) ✓ [3 marks]	**kph**

[3]

Past paper

Questions continue on the following page

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9. Charu needs to estimate the curved surface area of a hemisphere.

Charu knows:

- the radius of the hemisphere is 9 cm
- $\pi = 3.14159$

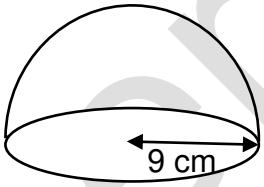
Charu rounds π to the nearest whole number to estimate the curved surface area.

What answer should Charu get?

Use:

$$A = \frac{4\pi r^2}{2}$$

Where:
A = curved surface area of sphere
r = radius of hemisphere



Not drawn to scale

Show your working.

Step 1: Formula for curved surface area of hemisphere:

Area = 4 × pi × r² ÷ 2 = 2 × pi × r²

Step 2: Substitute r = 9 cm and estimate pi ≈ 3 (or 3.14):

Area = 2 × 3 × 9² = 6 × 81 = 486 cm²

Answer 486 [3 marks] **cm²**

[3]

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10. Drew needs to order 200 paper cups for a party.

Drew needs small, medium and large cups in the ratio 14 : 4 : 2

Drew thinks there will be 108 more small cups than large cups.

Is Drew correct?

Show how you decide.

Step 1: Total parts in ratio = 14 + 4 + 2 = 20 parts

Step 2: Number of cups per part = 200 ÷ 20 = 10 cups

Step 3: Small cups = 14 × 10 = 140 | Large cups = 2 × 10 = 20

Step 4: Difference = 140 - 20 = 120 more small cups than large

120 ≠ 108. Therefore Drew is not correct (No).

Answer	No (Difference is 120) ✓ [3 marks]

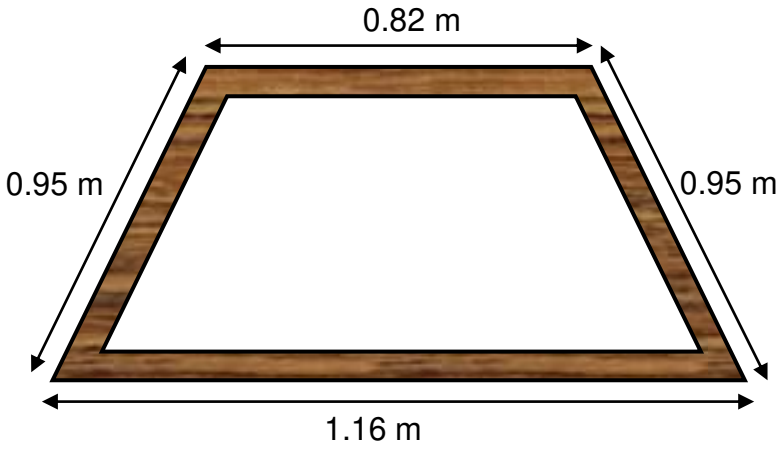
[3]

Past paper

Questions continue on the following page

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11. Kai wants to make a wooden picture frame with the dimensions shown below:



Not drawn to scale

Kai wants to buy 14% more wood than needed to allow for any mistakes.

Kai buys 4.5 metres of wood.

Has Kai bought enough wood?

Show your working.

Step 1: Perimeter of frame = 0.95 + 0.95 + 0.82 + 1.16 = 3.88 m

Step 2: Calculate with 14% extra wood:

3.88 × 1.14 = 4.4232 m

Step 3: Compare with 4.5 m available:

4.4232 m < 4.5 m. Kai has bought enough wood. Yes.

Answer Yes (4.42 … 4.5) [4 marks]

[4]

Past paper

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12. Billie owns a café. Deepal is the café manager.

The table shows how many customers the café had over two weeks.

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	22	38	46	30	44
Week 2	23	40	45	42	20

Deepal thinks that on average there were more customers per day on week 1.

Billie disagrees.

Show how **both** could be correct.

Show your working.

Step 1: Calculate Mean customer count:

Week 1 Mean = (22 + 38 + 46 + 30 + 44) ÷ 5 = 180 ÷ 5 = 36

Week 2 Mean = (23 + 40 + 45 + 42 + 20) ÷ 5 = 170 ÷ 5 = 34

Step 2: Calculate Median (middle value when ordered):

Week 1: 22, 30, 38, 44, 46 --> Median = 38

Week 2: 20, 23, 40, 42, 45 --> Median = 40

Give reason here:

- **If using the mean: Deepal is correct (Week 1 mean 36 > Week 2 mean 34).**
- **If using the median: Billie is correct (Week 2 median 40 > Week 1 median 38).**

Both are mathematically valid measures of average.

[4 marks]

[4]

Past paper

Questions continue on the following page

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13. Ashley wants to buy a motorbike for £9750

Ashley paid £5000 into a new savings account two years ago.

The savings account paid 4% compound interest per year.

Ashley will use all the money in the savings account as a deposit towards the motorbike.

The rest will be paid in 40 equal monthly payments.

How much will each monthly payment be?

Step 1: Savings after 2 years at 4% compound interest:

Year 1: £5000 × 1.04 = £5200

Year 2: £5200 × 1.04 = £5408 (or £5000 × 1.04^2)

Step 2: Remaining balance after deposit:

Remaining = £9750 − £5408 = £4342

Step 3: Monthly payment across 40 equal payments:

Monthly = £4342 ÷ 40 = £108.55

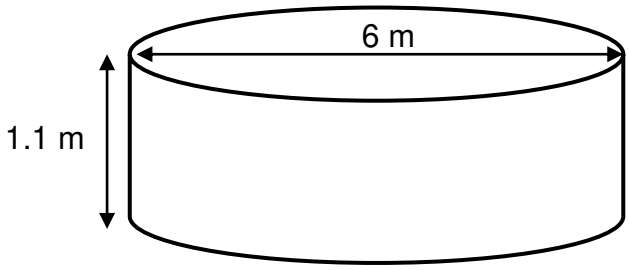
Answer £ 108.55 *[4 marks]*

[4]

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14. Taylor has a fishpond in the shape of a cylinder.



Not drawn to scale

Taylor uses this formula to work out how many fish can be put into the pond.

$$\text{Maximum number of fish} = \frac{\text{volume (in m}^3\text{) of pond multiplied by 220}}{1000}$$

What is the maximum number of fish Taylor can put in the pond?

Use $\pi = 3.14$

Step 1: Pond Volume = pi × r^2 × h

Diameter = 6 m --> Radius = 3 m, Depth = 1.1 m

Volume = 3.14 × 3^2 × 1.1 = 3.14 × 9 × 1.1 = 31.086 m^3

Step 2: Apply formula for maximum fish:

Fish = (31.086 × 220) ÷ 1000 = 6.83892 fish

Step 3: Whole number of fish (must round down): 6 fish

Answer	6 ✓ **fish**

[5 marks]

[5]

Questions continue on the following page

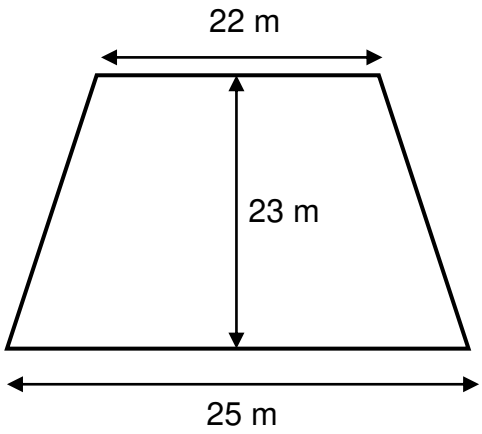
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15. Amory has turned a barn into an event venue.

Amory wants to estimate how much money the venue can be rented out for.

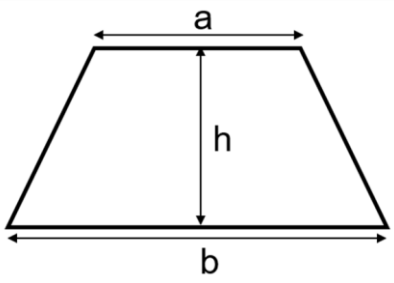
The floor plan of the venue is shown below:



Not drawn to scale

Use:

$$Area = \frac{(a+b)h}{2}$$



Amory estimates that one person will require 2.6 m^2 of floor space.

Amory finds these prices other local venues charge.

Venue	A	B	C	D	E
Price per person	£20	£35	£45	£30	£50

Amory will charge the median price.

What is the maximum amount of money Amory can expect to make?

Show your working.

Answer box is on the following page

Past Paper

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Step 1: Floor area of venue (Trapezium formula):

Area = ((a + b) ÷ 2) × h = ((22 + 25) ÷ 2) × 23

Area = 23.5 × 23 = 540.5 m²

Step 2: Capacity at 2.6 m² per person:

Capacity = 540.5 ÷ 2.6 = 207.884... people

Must round down to whole number: 207 people

Step 3: Median ticket price of local venues:

Prices: £20, £35, £45, £30, £50

Ordered prices: £20, £30, £35, £45, £50 --> Median = £35

Step 4: Maximum expected revenue:

Revenue = 207 people × £35 = £7245

Answer	£ 7245 *[6 marks]*

[6]

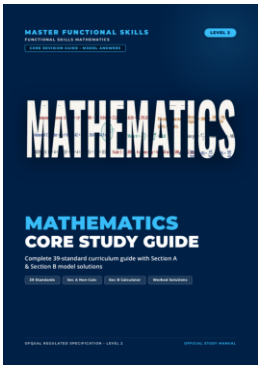
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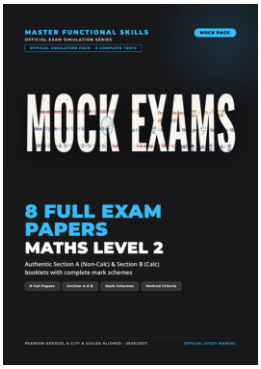


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